

## Instructions

1. This document outlines the tasks that you need to complete for this assessment item. Carefully read all sections of this document before attempting the assessment task.
2. This document must be read in conjunction with the general instructions and guidelines stated on the [assessment task Canvas page](#).
3. The UTS and Faculty of Science [assessment procedures and advice](#) also apply to this assessment task, so make sure you are familiar with these guidelines. The linked page provides general information about special consideration for assessments. Note that late submission penalties do not apply to this assessment since the only deliverable output is the oral presentation.
4. In your presentation, you must answer the questions detailed in the following pages. You may use PDF or Powerpoint slides for your presentation. *You must send your presentation slides via e-mail to [LenPatrickDominic.Garces@uts.edu.au](mailto:LenPatrickDominic.Garces@uts.edu.au) before your allocated time slot (ideally the day before).* Due to the IT security settings in place, it may not be possible to load your presentation slides on the computer in the room with a flash drive. Alternatively, you may bring your own device as long as it is HDMI or USB-C compatible.
5. You will be given only ten (10) minutes to present your responses. The teaching staff may terminate your presentation at the end of the allocated time.
6. If you have any questions about the assessment, please post these to the subject discussion forum on Canvas. If the enquiry relates to a personal matter, please contact the subject coordinator via e-mail.

## Mechanics

1. Before your presentation, you will roll a standard six-sided die to determine which question you will answer. If the result is an odd number, you will answer Question 1; otherwise, you will answer Question 2
2. You will then have ten (10) minutes to present your responses to the question. The teaching staff may terminate your presentation at the end of the allocated time if you have not yet finished your presentation.
3. The remainder of the time will be allocated to follow-up questions from the teaching staff. Follow-up questions will mainly be around your conclusions, the reasoning or method you used to reach those conclusions, and additional insights arising from your work.
4. This assessment is worth **[45 marks]** and accounts for 45% of your final mark in the subject. Each question is worth **[30 marks]**, the distribution of which is provided in the statement of the question. The remainder of your mark for this assessment will be based on:
  - Response to follow-up question/s **[12 marks]**
  - Referencing and GenAI usage disclosure **[3 marks]**

## Question 1: Portfolio Performance and Risk Measures

Consider an investment universe consisting of 30 risky assets (the constituents of the Dow Jones Industrial Average) and a risk-free asset whose return is  $r_0 = 1\%$  (per annum). Daily closing prices for these assets from 2 January 2015 to 31 December 2025 are provided in the file `DJIAShocksClosing.csv`. For simplicity, use an annualization factor of 252, reflecting the assumption that there are 252 trading days per calendar year.

1. Determine the mean-variance optimal portfolio with no risk-free allocation for the (annualized) target return of 25%. Report the allocation to the Walmart (WMT) stock.

[2 marks]

2. Calculate the Sharpe ratio, Jensen index, Treynor index, Sortino ratio, and diversification ratio for the mean-variance optimal portfolio from the previous question. Compare these results to those for a portfolio with equal weights in each of the 30 risky assets. Discuss your results, highlighting also the mean and standard deviation of returns of each portfolio.

[18 marks]

*The notation for the questions below follows the notation and assumptions in the lecture slides titled “Performance and Diversification Indicators and Risk Measures.” Note that these questions do not require the information or output from the implementation questions above. When presenting your solution, you only need to highlight the key steps in your oral delivery; the slides should display the full proof.*

3. Let  $\varphi_1$  and  $\varphi_2$  be two investment portfolios with nonzero risky allocations and let  $\varphi_\tau$  denote the tangent portfolio. Let  $\rho_{i,\tau} := \text{Cor}(\mathcal{R}(\varphi_i), \mathcal{R}(\varphi_\tau))$ , for  $i = 1, 2$ , and assume that  $\rho_{1,\tau} = \rho_{2,\tau}$ . Show that if  $\text{TI}(\varphi_1) > \text{TI}(\varphi_2)$ , then  $\text{SR}(\varphi_1) > \text{SR}(\varphi_2)$ .

[4 marks]

4. (Use the same notation as the previous sub-question) Suppose now that  $0 < \sigma_{\varphi_2} < \sigma_{\varphi_1}$  and  $\rho_{2,\tau} > \rho_{1,\tau}$ . Show that if  $\text{SR}(\varphi_1) > \text{SR}(\varphi_2)$ , then  $\text{JI}(\varphi_1) > \text{JI}(\varphi_2)$ .

[6 marks]

## Question 2: The Merton Problem

We consider a finite-horizon optimal consumption and investment problem using the framework that we have adopted in class. Given the probability space  $(\Omega, \mathcal{F}, \mathbb{P})$ , we assume that the investor's wealth evolves according to the SDE

$$dX(t) = (rX(t) + \theta(t)(\mu - r) - c(t)) dt + \theta(t)\sigma dW(t), \quad X(0) = x,$$

where  $x > 0$  is the investor's initial wealth,  $W$  is a standard  $\mathbb{P}$ -Brownian motion,  $r, \mu, \sigma$  are positive constants,  $\theta(t)$  is the amount of wealth invested in the risky asset, and  $c(t)$  is the consumption rate. Let  $T > 0$  be the time horizon and let  $\mathcal{A}(t, x)$  denote the set of admissible pairs  $(\theta, c) := \{(\theta(s), c(s))\}_{s \in [t, T]}$  for current wealth  $X(t) = x$ . Let  $\rho = (\mu - r)/\sigma$  denote the market price of risk.

The investor's objective is to maximize their expected utility from consumption and terminal wealth, i.e. their value function is

$$V(t, x) := \sup_{(\theta, c) \in \mathcal{A}(t, x)} \mathbb{E}^{t, x} \left[ \int_t^T h(s) U(c(s)) ds + AU(X^{\theta, c}(T)) \right],$$

where  $\mathbb{E}^{t, x}[\cdot]$  is the conditional expectation under  $\mathbb{P}$  given that  $X(t) = x$ ,  $h(t)$  is a given strictly positive function,  $A$  is given positive constant, and  $U$  is a utility function.

*Wherever applicable, when presenting your solution, you only need to highlight the key steps in your oral delivery; the slides should display the full proof.*

1. State the Hamilton-Jacobi-Bellman (HJB) equation corresponding to this stochastic optimal control problem.

[4 marks]

2. In terms of the partial derivatives of  $V$ , the model parameters, and the function  $h(t)$ , show the main steps in the derivation of the optimal consumption and investment strategies denoted by  $c^*(t, x)$  and  $\theta^*(t, x)$ , respectively.

[7 marks]

3. Assume that the investor has a power utility function  $U(x) = x^{1-\eta}/(1-\eta)$ , where  $\eta \geq 0$  and  $\eta \neq 1$ . Under the ansatz  $V(t, x) = f(t)x^{1-\eta}/(1-\eta)$  for some yet-to-be-determined function  $f$  satisfying  $f(T) = A$ , show that the function  $f$  must satisfy the ordinary differential equation

$$0 = f'(t) + (1 - \eta) \left( r + \frac{\rho^2}{2\eta} \right) f(t) + \eta(h(t))^{\frac{1}{\eta}} (f(t))^{\frac{\eta-1}{\eta}}$$

$$A = f(T).$$

[15 marks]

4. Calculate the Arrow-Pratt coefficient of absolute risk aversion  $A(x)$  for the investor assuming they have a power utility function. Express the optimal investment strategy in terms of  $\theta^*(t, x)$ .

[4 marks]